

Indian Institute of Technology Indore

Mechatronics And Instrumentation Lab



2-DOF SMA-Actuation Based Bio-Inspired Spy Robot

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Academic Year: 2025

2. Abstract

This project presents a 2-DOF SMA-Actuation Based Bio-Inspired Spy Robot that uses Shape Memory Alloy (SMA) springs to achieve silent and compact pitch–yaw motion inspired by the natural head movements of snakes. Unlike conventional motor-driven mechanisms, SMA actuation provides smooth and lifelike angular motion without bulky mechanical components, making the system ideal for surveillance tasks that require minimal noise and small form factor. The robot is built using modular 3D-printed components connected through Cardan joints, enabling controlled two-axis directional scanning while the robot remains stationary.

The actuation is powered by MOSFET-based driver circuits, and an ESP32 microcontroller coordinates the heating pulses required for SMA contraction and relaxation. An integrated ESP32-CAM module provides real-time video transmission over Wi-Fi, allowing the robot to function as a compact, stationary spy or monitoring unit. Testing demonstrates stable 2-DOF actuation and smooth camera-based observation, validating the robot’s potential for silent indoor surveillance, inspection tasks, and educational demonstrations of soft robotic actuation.

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4. Introduction

4.1 2-DOF SMA-Actuation Based Bio-Inspired Spy Robot

Modern surveillance applications require compact robotic systems capable of performing precise 2-DOF angular motion while remaining silent and unobtrusive. Traditional motor-based actuation introduces mechanical noise, gear backlash, and increased system size, making it unsuitable for tasks where stealth and smooth motion are important.

Biological organisms such as snakes and owls achieve accurate pitch–yaw orientation control through naturally efficient mechanisms. Inspired by this, the 2-DOF SMA-Actuation Based Bio-Inspired Spy Robot uses Shape Memory Alloy (SMA) springs to generate controlled thermomechanical movement without motors or gears. SMA actuation provides silent operation, high compactness, and smooth kinematic response, making the robot well-suited for stationary surveillance, directional monitoring, and soft-robotics experimentation.

4.2 Why Shape Memory Alloy (SMA) Actuation

Shape Memory Alloy (SMA) springs contract when heated and return to their original length when cooled, enabling controlled pitch–yaw angular motion without the need for motors, gears, or bulky rotary mechanisms.

SMA-based actuation provides:

- **Silent Operation:** No motor noise, making it ideal for stealth and surveillance tasks.
- **Compactness:** SMA springs allow a small, lightweight design with minimal mechanical parts.
- **High Torque-to-Weight Ratio:** SMA contraction generates sufficient force for precise 2-DOF movements.
- **Modularity:** Each SMA actuator can be individually replaced or reconfigured.
- **Smooth Pitch–Yaw Motion:** Thermomechanical actuation produces natural, continuous angular movement.

The ESP32 microcontroller regulates SMA heating pulses and manages Wi-Fi-based video transmission through the ESP32-CAM module, enabling the robot to function as a compact and silent 2-DOF bio-inspired spy system.

5. Problem Statement

5.1 Problem Statement

Modern indoor surveillance and monitoring tasks require robotic systems that are compact, silent, and capable of precise directional scanning. However, most existing surveillance mechanisms face significant limitations:

- **Bulky Motor-Based Actuation:** Conventional pan-tilt systems rely on motors and gears, which increase size, weight, and mechanical complexity.
- **Mechanical Noise and Vibrations:** Motor-driven mechanisms generate audible noise and vibrations, making them unsuitable for stealth or sensitive monitoring environments.
- **High Power Consumption:** Continuous motor operation demands more electrical power, reducing efficiency and limiting compact battery-powered designs.
- **Limited Biomimetic Motion:** Traditional actuation mechanisms cannot replicate smooth, natural pitch-yaw movements required for subtle, lifelike surveillance behavior.

5.2 Our Solution

This project proposes a 2-DOF SMA-actuated bio-inspired spy robot designed for silent and precise directional surveillance. The system replaces conventional motors with SMA springs to achieve smooth pitch-yaw motion.

Key Solution Points:

- **Compact and Silent Design:** The robot uses SMA springs instead of motors, eliminating noise and reducing mechanical complexity.
- **2-DOF Pitch-Yaw Motion:** SMA-based thermomechanical actuation provides smooth, lifelike angular movement for directional scanning.
- **Motorless Actuation Mechanism:** No gears or rotary motors are required, resulting in a lightweight and low-maintenance structure.
- **Arduino Nano for Actuation Control:** The Arduino Nano generates PWM signals and triggers MOSFET drivers to heat the SMA springs.
- **ESP32-CAM for Wireless Monitoring:** The ESP32-CAM module handles Wi-Fi streaming and real-time surveillance.



3D Model

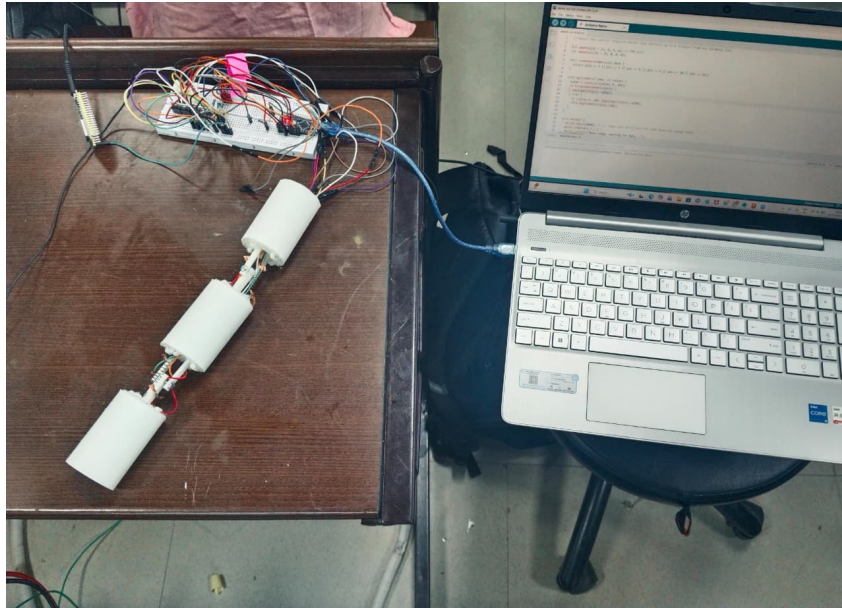
- **Modular 3D-Printed Structure:** Cardan-joint-based frame allows easy assembly, maintenance, and actuator replacement.

Key Features:

- **Silent SMA-Based Pitch–Yaw Motion:** Enables discreet surveillance without motor noise.
- **Compact 3D-Printed Mechanism:** Lightweight Cardan-joint structure allowing precise 2-DOF angular control.
- **MOSFET-Based SMA Drivers:** Efficiently switches high-current SMA heating cycles under Arduino control.
- **Arduino Nano Control Interface:** Generates timed PWM pulses and handles actuation commands for both SMA channels.
- **ESP32-CAM Wireless Streaming:** Provides live video output for real-time observation and monitoring.
- **Modular Design:** SMA springs, joints, and electronic components can be easily replaced or upgraded.

5.3 Advantages of the Solution

- **Silent Operation:** SMA-based thermomechanical actuation produces no motor noise, making the robot ideal for discreet surveillance tasks.
- **Compact and Lightweight Design:** The use of SMA springs and 3D-printed components results in a small, low-profile mechanism suitable for indoor monitoring.
- **Low Mechanical Wear:** With no gears, motors, or rotating shafts, the system experiences minimal friction and requires very little maintenance.

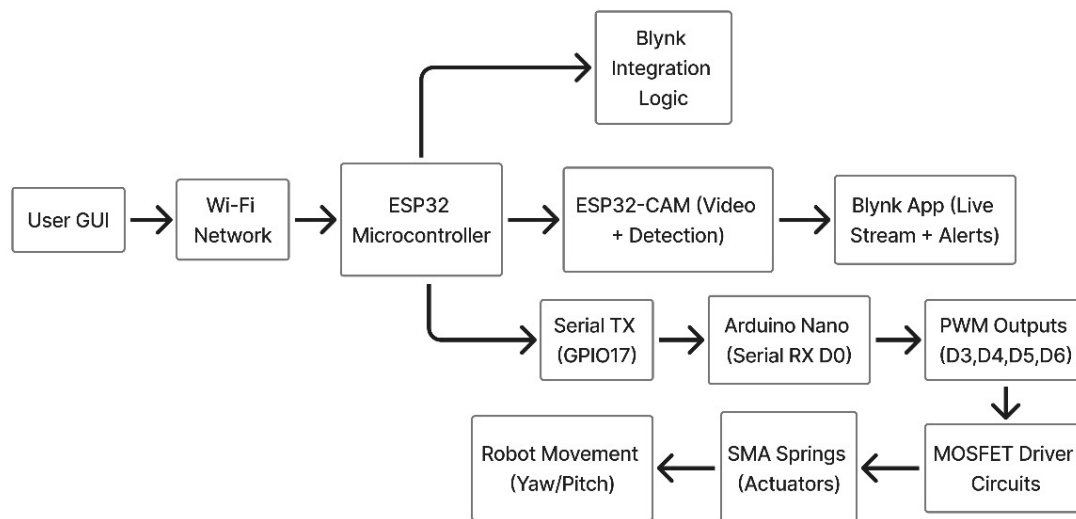


Entire Setup

- **Modularity and Easy Upgradability:** SMA actuators, 3D-printed joints, the Arduino Nano, and the ESP32-CAM module can be independently replaced or upgraded.
- **Smooth 2-DOF Motion:** SMA actuation enables fluid, lifelike pitch–yaw movement that mimics biological scanning behavior.
- **Efficient Control Architecture:** The Arduino Nano provides simple PWM control for SMA drivers, while the ESP32-CAM manages real-time wireless video, ensuring clear functional separation.
- **Low Power Minimal Components:** The motorless actuation and compact electronics reduce overall system power consumption and complexity.

6. System Design & Implementation

The robot integrates mechanical, electrical, and sensing components into a unified, modular architecture suitable for confined-space mobility. The design emphasizes robustness, long life, and ease of maintenance.



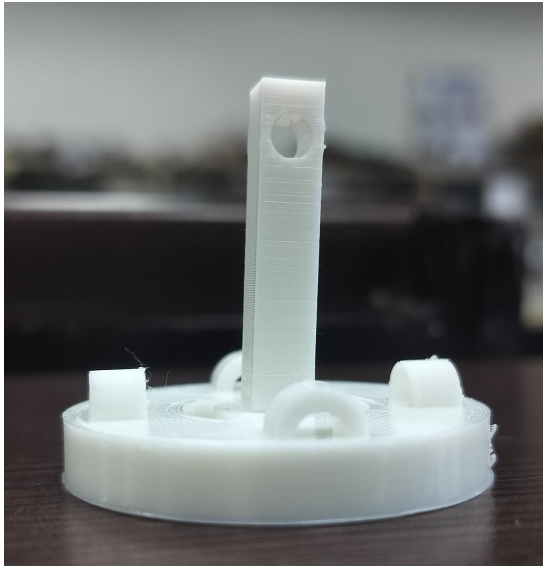
System Architecture

6.1 Mechanical Design

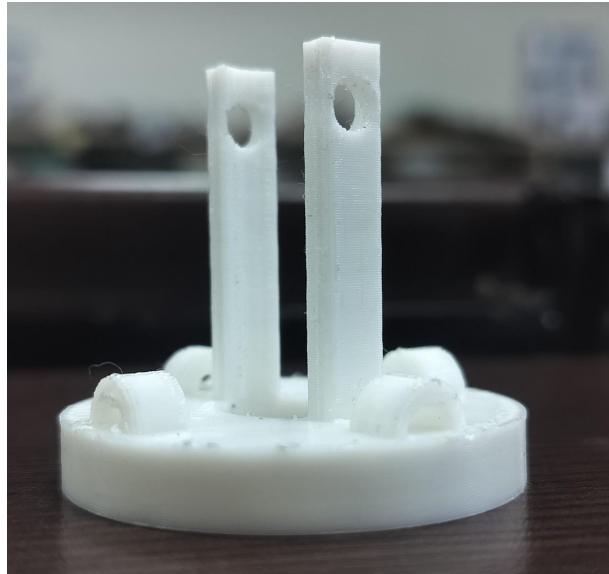
- Compact 3D-printed structure with a simple Cardan-style 2-DOF joint.
- Circular base with vertical hinge arms for pitch–yaw rotation.
- SMA springs attached between hinge arms and upper plate for actuation.

Key aspects:

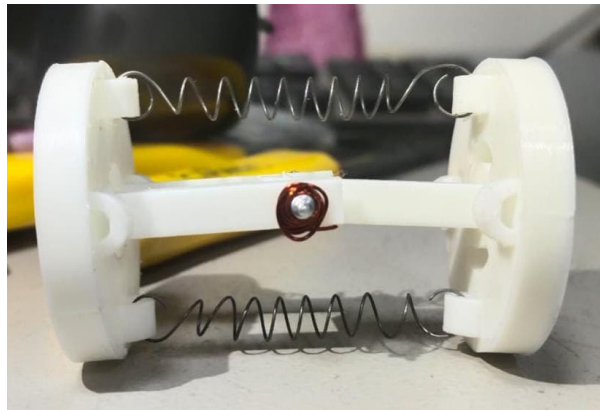
- **3D-Printed Construction:** lightweight and customizable.
- **Flexible Joints:** allow smooth motion in narrow pathways.
- **Modular Segments:** easy maintenance and future expansion.
- **Optimized SMA Layout:** ensures efficient movement generation.



Fixture 1 for hinge joint



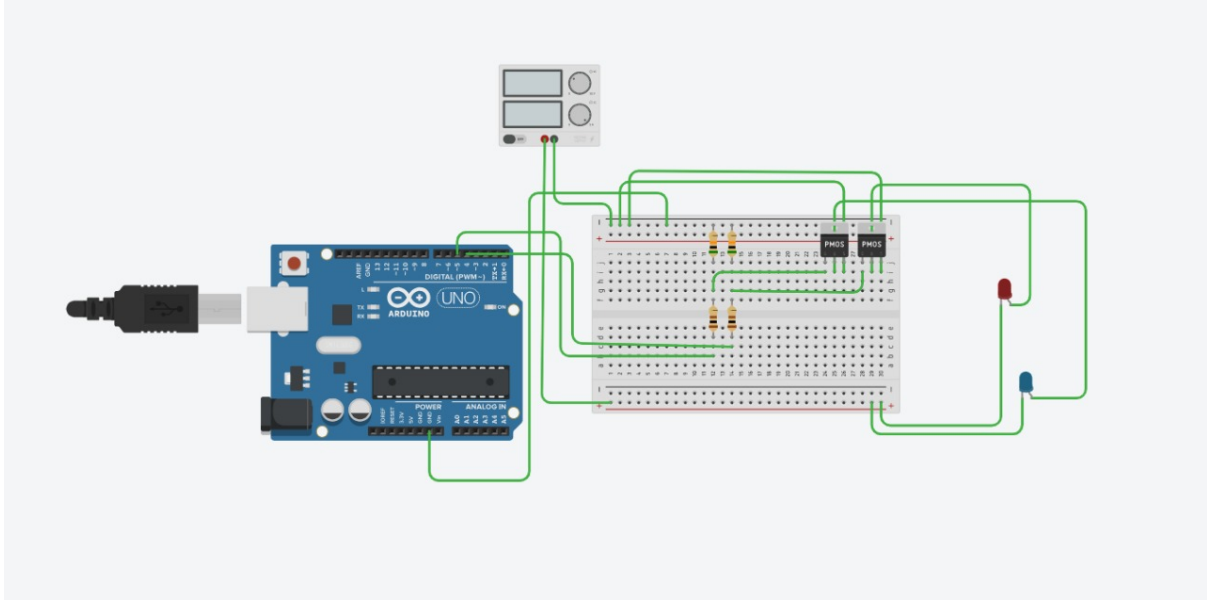
Fixture 2 for hinge joint



Fixtures Assembly

6.2 Electrical & Control Hardware

- **Arduino Nano Control Unit:** Generates PWM signals to drive the SMA springs through MOSFET switching circuits.
- **MOSFET Driver Circuits:** Regulate high-current SMA heating cycles for reliable pitch–yaw actuation.
- **ESP32-CAM Module:** Provides wireless real-time video streaming for surveillance.
- **Power Distribution System:** A regulated DC supply powers the Arduino Nano, MOSFET drivers, SMA actuators, and ESP32-CAM.

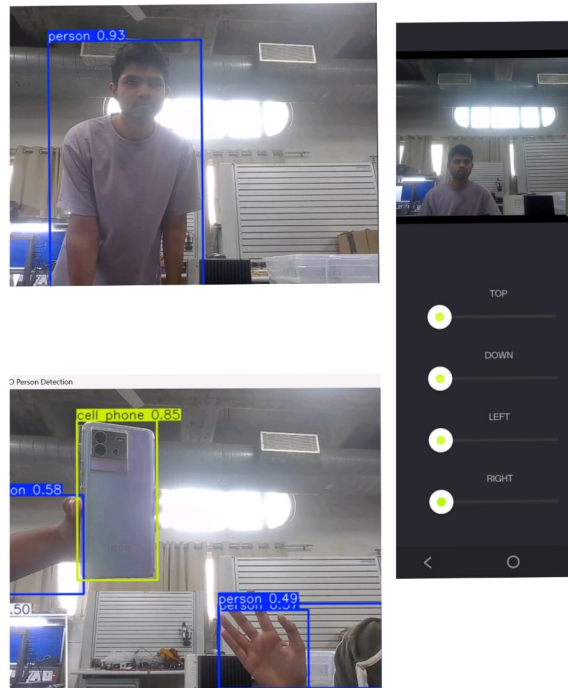


Electric Circuit

6.3 Working Principle

The robot's operation is a coordinated process integrating the software interface, a dual-microcontroller architecture, and the mechanical actuators. The control flow, as detailed in the System Architecture diagram, is as follows:

- **User Input:** The user controls the robot via the **Blynk mobile application**. Four independent sliders on the GUI allow for precise manual control over each of the four SMA springs (SMA 1–4). These commands are sent over a Wi-Fi network.
- **Main Processing (ESP32):** The **ESP32 microcontroller** serves as the primary brain. It receives slider commands from Blynk and also manages the **ESP32-CAM** module, which provides a live video feed and runs a **YOLO-based human detection** algorithm.
- **Motion Command:** The ESP32 translates the desired movement (pitch/yaw) from the slider inputs into high-level commands. It then sends these commands via a **Serial (TX)** connection to the Arduino Nano, which functions as a dedicated actuator controller.
- **PWM Generation (Arduino Nano):** The **Arduino Nano** receives the serial commands and generates four independent **PWM (Pulse Width Modulation) signals** (on pins D3, D4, D5, D6) to control the precise heating cycle for each SMA spring.
- **Power Switching (MOSFETs):** The low-current PWM signals from the Nano are fed into **MOSFET driver circuits**. These MOSFETs act as high-current electronic switches, regulating the power flow from the main power distribution system to the SMA springs.
- **Mechanical Actuation:** When a MOSFET is triggered, it allows a high current to pass through its corresponding **SMA spring**. The spring heats up and **contracts** (like a muscle). By selectively contracting one or two of the four springs, a pulling force is exerted on the **2-DOF Cardan joint**, causing the entire segment to bend and produce the desired ****pitch or yaw movement****.



YOLO and Software Interface

6.4 Software Interface

The software control system for the robot is implemented using the Blynk mobile application, providing a simple and user-friendly interface to operate the SMA actuators and monitor the live camera feed. The interface includes four slider widgets assigned to SMA 1, SMA 2, SMA 3, and SMA 4, allowing precise manual control of the pitch–yaw motion by adjusting the heating intensity of each SMA spring. Additionally, a YOLO-based human detection algorithm is integrated into the system to automatically identify human presence in the camera feed. When a person is detected, the system triggers an alarm notification through the Blynk app for immediate alert.

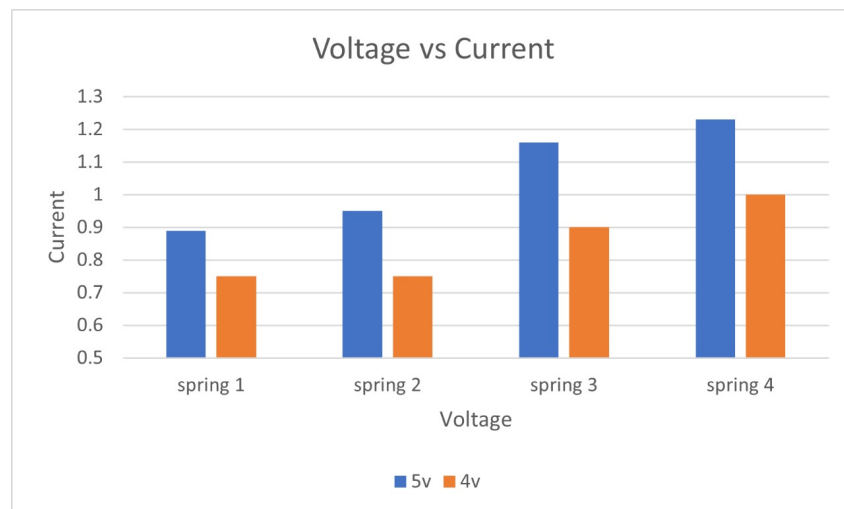
- **SMA Control Sliders:** Four independent sliders in Blynk control SMA 1–SMA 4 for smooth and adjustable actuation.
- **Live Video Stream:** The ESP32-CAM module streams real-time video to the Blynk app for continuous remote monitoring.
- **Wireless Operation:** All commands and data are transmitted over Wi-Fi, enabling completely cable-free control and monitoring.
- **YOLO-Based Human Detection:** A lightweight YOLO model processes the video feed to detect human presence in real-time.
- **Automatic Alarm Notification:** When YOLO detects a human, the system immediately sends an alert message or buzzer notification through the Blynk dashboard.

7. Testing Results

To validate the robot's pitch and yaw capabilities, a series of tests were conducted. These tests focused on characterizing the electrical properties of the SMA actuators, measuring their maximum angular deflection, and determining the optimal operating frequency. All tests were conducted using controlled PWM signals to ensure repeatable, safe operation.

7.1 Actuator Electrical Characterization

Before implementing control logic, each of the four SMA springs was tested to understand its voltage-current (V-I) relationship. This is critical for ensuring the power supply can meet the demand and for identifying any manufacturing inconsistencies between the actuators. A variable power supply was used to apply 4V and 5V to each spring, and the resulting current draw was measured. The results, shown in Figure ??, demonstrate:



Voltage vs. Current draw for each of the four SMA springs.

- **Current at 4V:** The current ranged from approximately 0.75 A (Springs 1 2) to 0.9 A (Spring 3).
- **Current at 5V:** The current ranged from approximately 0.9 A (Spring 1) to over 1.2 A (Spring 4).

This test confirms that a full-strength actuation (using a 5V supply) operates near the expected 1 A range.

7.2 Pitch & Yaw Angular Deflection Tests

The primary mechanical test involved measuring the maximum angular deflection for pitch and yaw movements. A protractor was used to measure the robot segment's angle relative to its neutral 0° baseline.



Experimental measurement of pitch/yaw angles. Left image (Yaw): A maximum bend of $\sim 40^\circ$. Right image (Pitch): A maximum bend of $\sim 20^\circ$.

As shown in Figure ??, the tests yielded two key benchmarks:

- **Maximum Yaw Deflection:** By applying a full-strength signal (PWM 255, drawing $\sim 1A$) for horizontal movement (yaw), the segment achieved a maximum, repeatable bend of approximately 40° .
- **Maximum Pitch Deflection:** Using the same full-strength signal for vertical movement (pitch), a controlled bend of only 20° was consistently achieved.
- **Observation on Gravity:** A clear difference in deflection was observed between the two axes. The maximum pitch angle ($\sim 20^\circ$) is significantly less than the maximum yaw angle ($\sim 40^\circ$). This is attributed to the **effect of gravity**, as the pitch movement requires the actuator to lift the weight of the robot segment, while the yaw movement is horizontal and not directly opposed by gravity.

These real-world measurements confirm the robot's significant range of motion, while also highlighting the impact of gravitational forces on the actuators.

8. Limitations & Future Work

Although the 2-DOF SMA-actuated spy robot performs smooth pitch–yaw motion and reliable video streaming, several limitations remain that affect its performance and usability.

8.1 Current Limitations

- **Slow Cooling Response:** SMA springs require time to cool down before returning to their original shape, limiting the maximum achievable actuation speed.
- **Thermal Accumulation:** Continuous activation can cause residual heating in the SMA wires, reducing stroke range and affecting repeatability over long durations.
- **High Current Requirement:** SMA actuation demands relatively high current, which restricts battery-powered operation and requires proper MOSFET driving circuitry.
- **Limited Angular Precision:** SMA deformation is nonlinear, meaning fine-tuned pitch–yaw control can be difficult without additional sensing or feedback mechanisms.

8.2 Further Improvements

Using two-way SMA springs for forward motion allows the development of a compact, motorless snake-type robot suitable for earthquake search and rescue. When coordinated across multiple segments, two-way SMA elements (or antagonistic SMA pairs) can generate inch-worm, serpentine, or lateral-undulation gaits that enable the robot to move through narrow gaps and over debris.

- The robot can be extended into a snake-type rescue system capable of crawling through rubble and narrow gaps formed during earthquakes.
- Two-way SMA actuation enables forward motion, allowing the robot to move inside collapsed structures where human entry is not possible.
- Its compact and silent movement makes it suitable for operating in unstable debris zones without causing additional disturbance.
- With an ESP32-CAM and sensors, the robot can assist rescue teams by locating trapped survivors and transmitting real-time visual information.

9. Conclusion

This project demonstrates the successful development of a compact, silent, and bio-inspired 2-DOF spy robot actuated entirely using Shape Memory Alloy (SMA) springs. By replacing conventional motors with SMA-based thermomechanical actuation, the system achieves smooth pitch–yaw motion in an extremely lightweight and low-noise design. The integration of a 3D-printed Cardan-style joint, MOSFET-driven SMA control, and an ESP32-CAM module enables precise directional scanning along with real-time wireless video streaming.

Experimental evaluation confirmed stable angular motion, reliable SMA actuation, and clear camera performance, validating the robot’s suitability for applications such as indoor surveillance, inspection tasks, and soft-robotics research. The modular mechanical and electronic architecture allows easy replacement of SMA springs, control boards, and structural components, making the system cost-effective and easy to upgrade.

Future extensions of this platform include expanding the 2-DOF module into a multi-segment SMA-driven snake robot capable of forward locomotion for earthquake search and rescue. With additional sensing and enhanced locomotion strategies, this SMA-based design holds strong potential for evolving into a practical, adaptable, and biologically inspired robotic tool for confined-space exploration and post-disaster response.